

To Learn It, Teach It

Using Explanation, Retrieval, and the Feynman Technique to Build Real Understanding

August 16, 2026

Have you ever read a page, reached the bottom, and realized that you could not explain what you just read? The words looked familiar while they were in front of you. The sentences seemed to make sense. You may even have felt confident that you understood them. Yet, when the page was closed, the idea seemed to disappear.

This happens because recognizing information is not the same as understanding it. When the answer is visible, it is easy to mistake familiarity for knowledge. The real test comes when the book is closed, the notes are put away, and someone asks, “What does this mean?”

One of the best ways to discover whether you understand something is to try to teach it to another person. Teaching forces you to do more than repeat a definition. You must decide what is important, arrange the ideas in a sensible order, choose words another person can understand, and answer questions. Somewhere in that process, the truth becomes difficult to avoid: either you understand the idea well enough to explain it, or you have found the part you still need to learn.

That discovery is the foundation of a learning method commonly called the **Feynman Technique**.

Understanding Instead of Memorizing

Richard Feynman was a Nobel Prize-winning physicist who became famous for explaining difficult subjects with unusual clarity. Although the familiar step-by-step “Feynman Technique” appears to have been organized and named by later writers, it reflects something important about the way Feynman approached knowledge. He did not believe that knowing the name of something was the same as understanding it.

Technical vocabulary can sometimes create an illusion of knowledge. A person may repeat an impressive definition without knowing what it means. If asked a slightly different question, the memorized answer no longer works. True understanding is more flexible. It allows a person to describe the same idea in different ways, connect it to an example, and apply it to a new situation.

Imagine that you memorize the statement, “A strong password helps prevent unauthorized access.” You may be able to repeat that sentence on a quiz. But do you understand it well enough to explain why a long password is generally safer than a short one? Could you explain why using the same password for every account is dangerous? Could you help a younger student create a safer password without simply repeating the original sentence?

Those questions move beyond memorization. They require you to use the idea.

Feynman believed that learning required students to discuss ideas, think about them, and work through them personally. He warned against assuming that hearing a good explanation was enough. A teacher can guide your thinking, but the teacher cannot do the thinking inside your mind. At some point, you

must take possession of the idea and make sense of it for yourself. Feynman expressed this view in the preface to *The Feynman Lectures on Physics*.

Why Teaching Works

Teaching changes the way you approach a subject. If you study only because you expect a quiz, you may concentrate on remembering the exact words likely to appear on it. If you expect to teach the subject, you begin looking for meaning. You ask different questions: What is the main idea? Which part should come first? What example would make this easier to understand? What might confuse someone? How are the pieces connected?

Educational research supports this effect. In one study, students who expected to teach the material later remembered more important information and organized it more effectively than students who expected only to take a test. The expectation of teaching changed how they learned from the beginning. The research was published in *Memory & Cognition*.

Teaching also requires you to retrieve information. When you explain something with your notes closed, you must bring the idea out of memory instead of simply looking at it again. This is more difficult than rereading, but the difficulty is part of what makes it useful. Each attempt to retrieve and organize the idea strengthens your ability to remember and use it later.

Researchers sometimes call the benefits of learning through teaching the **protégé effect**. A protégé is someone you guide or teach. Studies have found that students may work harder and pay closer attention when they feel responsible for another learner's success. If the learner becomes confused, the student-teacher must reconsider the explanation and improve it. Research from Stanford describes how students learned by teaching a computer-based pupil.

The listener benefits from receiving an explanation, but the teacher benefits from having to construct one.

Begin with Something Specific

The Feynman Technique begins by choosing something you want to understand. It should be specific enough to explain in a few minutes. "Computers" would be much too broad. "Why should I avoid reusing the same password?" would be manageable. Outside cybersecurity, you might choose "Why do seasons change?" "How does a bill become a law?" or "What makes a story's narrator unreliable?"

Study the subject normally at first. Read, watch, listen, take notes, or examine examples. However, study with the expectation that you will soon have to explain the idea without depending on the original source.

That expectation matters. Instead of copying every sentence, look for the structure underneath the information. Try to identify the central idea, the supporting details, and the connections between them. Ask yourself what someone must understand first before the rest will make sense.

Once you have studied, put the source away.

Explain It in Your Own Words

Now teach the idea. You can explain it to a classmate, a friend, a parent, or a younger student. If no one is available, you can teach an imaginary student, speak to an empty chair, make an audio recording, or even explain it to a pet. The listener does not need to understand the subject beforehand. In fact, explaining it to a complete beginner may be especially useful because you cannot rely on unexplained vocabulary.

Suppose you are teaching someone why suspicious messages can be dangerous. You might begin by explaining that a dishonest person can make a message look as if it came from someone trustworthy. The message may try to create fear, excitement, or urgency so that the reader clicks a link or shares private information without stopping to think.

That explanation is more useful than simply repeating the word *phishing*. It describes what happens and why it works. Once the basic idea is clear, the correct vocabulary can be added. The word now represents an idea the learner understands instead of acting as a substitute for understanding.

As you explain, avoid copying the textbook's sentences from memory. Use your own words. Create your own example. If possible, explain the idea in more than one way. Being able to change the explanation shows that you understand the meaning rather than only the wording.

Find the Place Where the Explanation Breaks

Sooner or later, you will reach a difficult spot. You may forget what happens next. You may use a word that you cannot define. You may realize that your example does not quite match the idea. Your listener may ask a simple question that you cannot answer.

This can feel embarrassing, but it is actually the most valuable moment in the process. You have found a gap in your understanding.

The goal is not to hide that gap or talk around it. Stop and name it. You might say, "I understand why the message looks convincing, but I am not sure how someone can make the sender's identity look real." That sentence turns a vague feeling of confusion into a specific question you can investigate.

Return to the book, lesson, demonstration, or another reliable source. Look for the missing explanation. Check whether the rest of your understanding was correct. Then close the source and teach the idea again.

Learning through the Feynman Technique is therefore not a straight line. It is a cycle:

Study. Explain. Find the gap. Investigate. Explain again.

Each time through the cycle, the explanation becomes clearer and your understanding becomes stronger.

Make It Simple, but Keep It True

A good explanation uses language appropriate for the listener. If a younger student does not know a technical word, define it or replace it temporarily with familiar language. Comparisons and examples can help someone form a mental picture of an unfamiliar idea.

However, simple does not mean careless. An explanation should not become so simple that it is no longer accurate. If your comparison works only partly, explain where it stops working. If the idea has an exception, acknowledge it. The goal is to remove unnecessary confusion without removing the truth.

This balance is one of the most valuable skills a learner can develop. It is easy to make an explanation complicated when you are repeating words from a source. It is also easy to make an explanation simple by leaving out everything difficult. Real understanding allows you to make an idea both clear and accurate.

Let the Other Person Ask Questions

Questions are a gift to the person teaching. They test parts of the explanation that you may not have considered.

If you explain why a password should be kept private, a listener might ask, “What if I share it only with my best friend?” If you explain why a surprising message should be treated carefully, someone may ask, “What if it really does come from my teacher?” These questions require more than a memorized rule. You must explain the reasoning behind the rule.

A question you cannot answer is not proof that the teaching attempt failed. It shows you what to study next. A good learner becomes comfortable saying, “I don’t know yet, but I know what I need to find out.”

This is a much stronger response than pretending to understand.

Teaching Is Part of Learning

You do not have to wait until you are an expert before teaching. Teaching can be one of the ways you become more knowledgeable. The important thing is to be honest about what you know, verify uncertain information, and remain willing to correct your explanation.

This practice will matter in almost any career. People constantly teach coworkers, customers, family members, and friends. In cybersecurity, a professional may need to explain why a message is dangerous, why a safety rule matters, or what someone should do after making a mistake. The explanation must be understandable to people who do not have technical training.

Yet the value of the Feynman Technique is much larger than any one career. It is a way of becoming an independent learner. It helps you move beyond filling in blanks, repeating definitions, and memorizing answers long enough to pass a test. It teaches you to examine your own thinking.

The next time you believe you understand something, try teaching it. Put away the notes and begin with the simplest question: “What is this really about?”

If the explanation comes easily, test it with examples and questions. If it falls apart, do not be discouraged. You have not discovered that you are incapable of learning. You have discovered exactly where the learning needs to continue.

That is progress—and it is one of the reasons teaching someone else can be one of the best ways to teach yourself.